

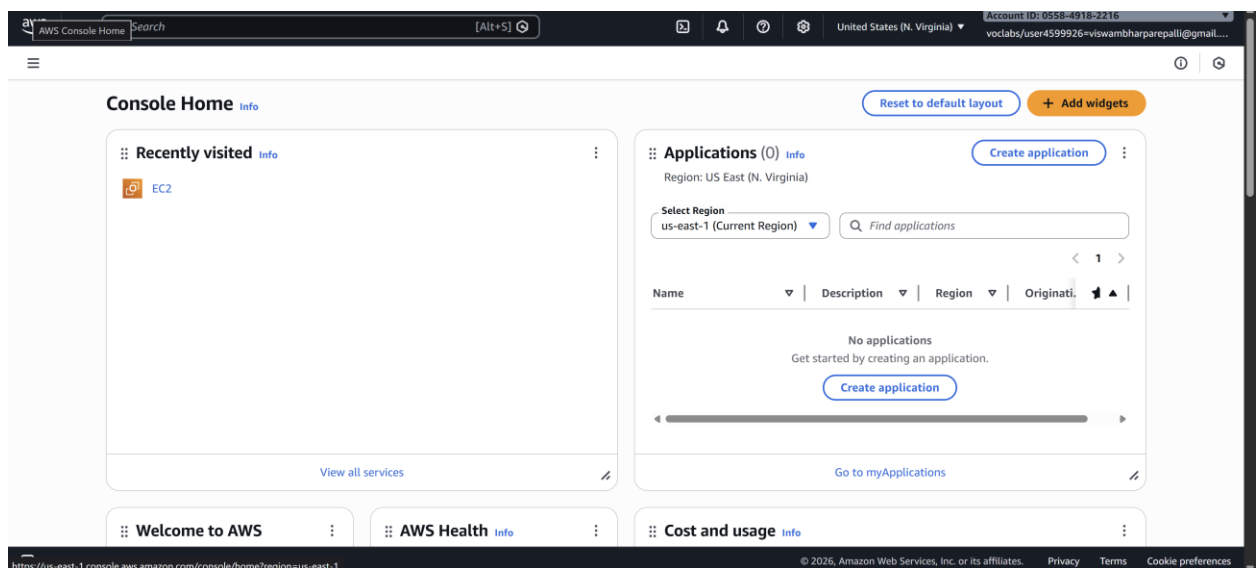
Week - 1

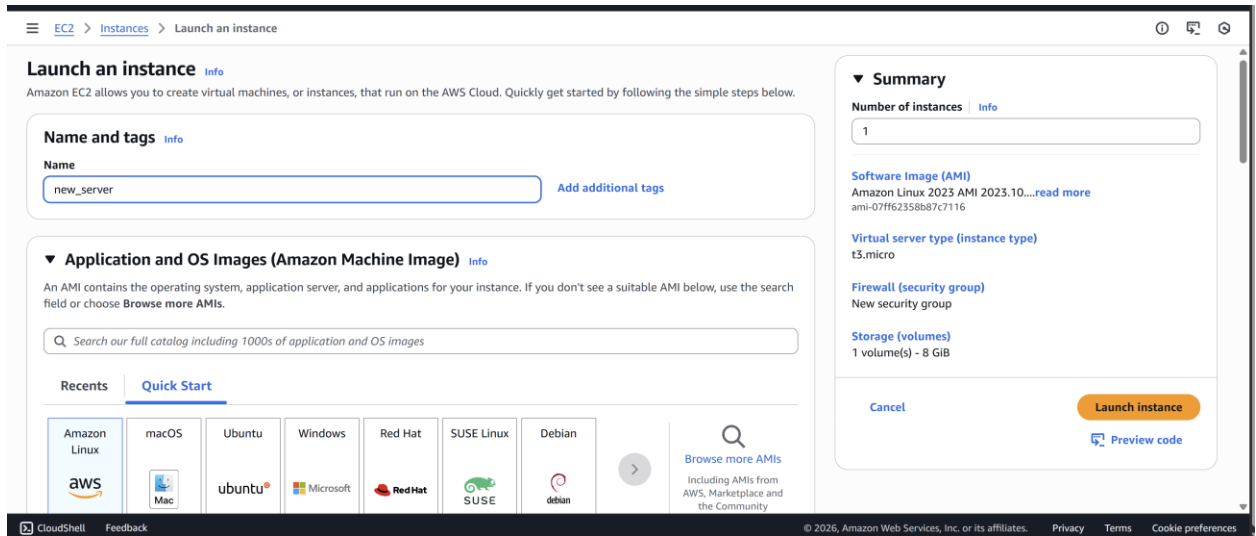
1. Creation of Amazon EC2 Instances (Linux and Windows) and Establishing Secure Connections Using SSH from Local Machine, PuTTY (.pem & .ppk), AWS CloudShell, and Remote Desktop Protocol (RDP)

□ PREREQUISITES

Make sure you have:

- ☒ EC2 **Ubuntu** instance running
- ☒ .ppk private key file
- ☒ PuTTY installed (Windows)-- <https://putty.org/index.html> (INSTALL FIRST ONE)





◆ STEP 1: Check Security Group

Go to **EC2 → Security Groups → Inbound rules**

Ensure this rule exists:

Type: SSH

Port: 22

Source: My IP

✗ If SSH is blocked → connection will fail

Create key pair

Key pair name

Key pairs allow you to connect to your instance securely.

new_key1

The name can include up to 255 ASCII characters. It can't include leading or trailing spaces.

Key pair type

☒ RSA
RSA encrypted private and public key pair

☐ ED25519
ED25519 encrypted private and public key pair

Private key file format

☐ .pem
For use with OpenSSH

☒ .ppk
For use with PuTTY

⚠️

When prompted, store the private key in a secure and accessible location on your computer. You will need it later to connect to your instance. [Learn more](#)

Cancel

Create key pair

aws

Search

[Alt+S]

United States (N. Virginia)

Account ID: 0558-4918-2216

EC2

Instances

i-0c30302062262a5d5

EC2

Dashboard

EC2 Global View

Events

Instances

Instance Types

Launch Templates

Spot Requests

Savings Plans

Reserved Instances

Dedicated Hosts

Capacity Reservations

Capacity Manager

Images

AMIs

AMI Catalog

Elastic Block Store

Instance summary for i-0c30302062262a5d5 (new_server)

Updated less than a minute ago

Instance ID

i-0c30302062262a5d5

IPv6 address

-

Hostname type

IP name: ip-172-31-19-52.ec2.internal

Answer private resource DNS name

IPv4 (A)

Auto-assigned IP address

13.222.128.215 [Public IP]

IAM Role

-

IMDSv2

-

Public IPv4 address

13.222.128.215 | open address

Instance state

Running

Private IP DNS name (IPv4 only)

ip-172-31-19-52.ec2.internal

Instance type

t3.micro

VPC ID

vpc-023fdd5a5e0711563

Subnet ID

subnet-0917b82aaaa230c35

Instance ARN

-

Private IPv4 addresses

172.31.19.52

Public DNS

ec2-13-222-128-215.compute-1.amazonaws.com | open address

Elastic IP addresses

-

AWS Compute Optimizer finding

Opt-in to AWS Compute Optimizer for recommendation

Auto Scaling Group name

-

Connect

Instance state

Actions

https://us-east-1.console.aws.amazon.com/console/home?region=us-east-1

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STEP 2: Open PuTTY

1. Launch PuTTY

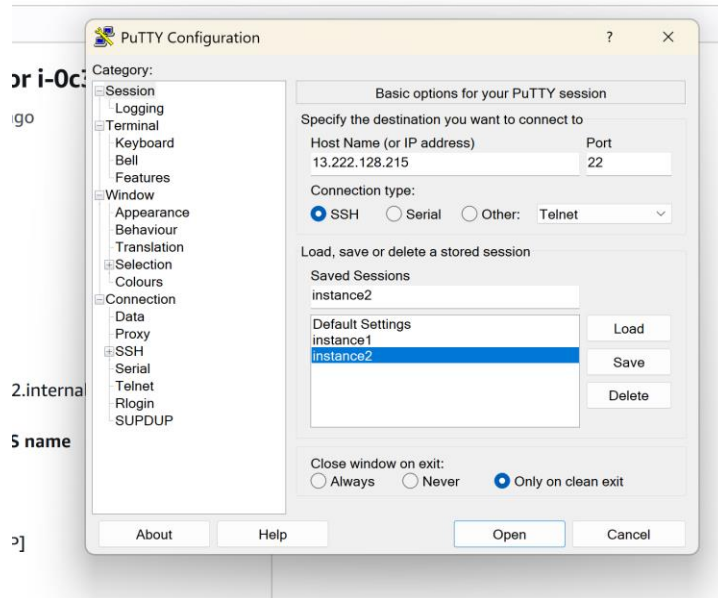
2. In **Session** screen:

Host Name (or IP address): EC2—CONNECT—COPY (DNS NAME)

ubuntu@<PUBLIC-IP>

Example:

[ubuntu@3.110.xxx.xxx](#)

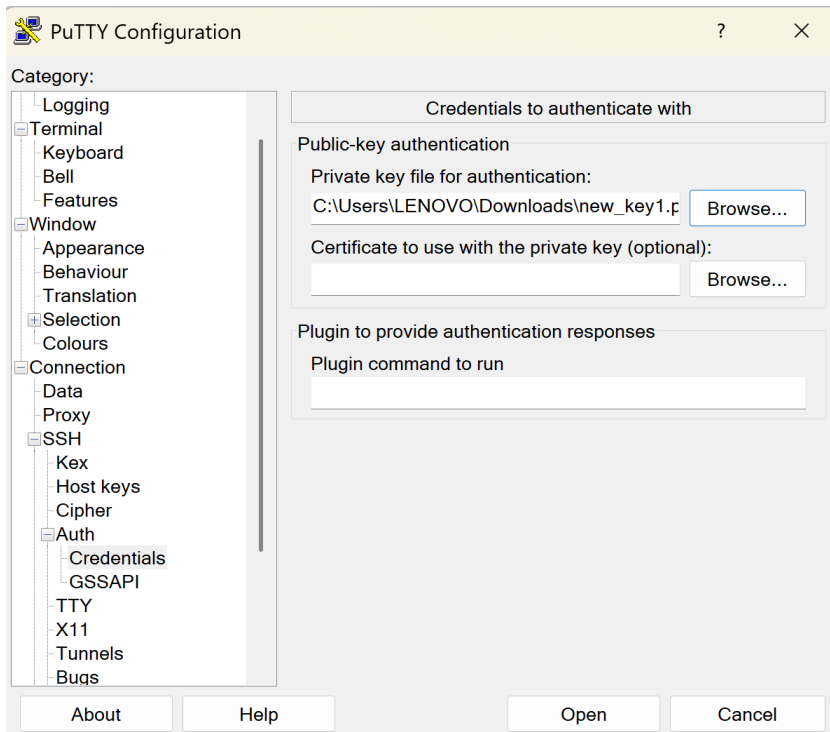


◆ STEP 3: Attach the .ppk Key

In PuTTY left menu, go to:

Connection → SSH → Auth → Credentials

1. Click **Browse**
2. Select your **.ppk file**
3. Click **Open**



◆ STEP 4: Connect to EC2

Click **Open**

- First time → click **Yes** (security alert)
- PuTTY will log in automatically as:

ubuntu

🔔 You are now connected to your Ubuntu EC2 instance

- ☒ **EC2 Ubuntu instance running**
- ☒ **Public IPv4 address** (or Elastic IP)
- ☒ **Key pair (.pem file)** used while launching EC2
- ☒ **PuTTY & PuTTYgen installed**

📄 Download PuTTY:

<https://www.putty.org>

💎 STEP 1: Convert .pem to .ppk using PuTTYgen

PuTTY cannot use .pem directly.

Steps:

1. Open **PuTTYgen**
2. Click **Load**
3. Select your .pem file
(change file type to *All Files*)
4. Click **Open**
5. Click **Save private key**
6. Save as keyname.ppk

✓ Conversion done

💎 STEP 2: Get EC2 Public IP

Go to:

AWS Console → EC2 → Instances---CONNECT –COPY DNS NAME

💎 STEP 3: Configure PuTTY

📄 Open PuTTY

Session:

- Host Name (or IP address): **dns name**
-

2 Load Private Key

Go to:

Connection → SSH → Auth → Credentials

- Click **Browse**
 - Select your .ppk file
-

◆ STEP 4: Connect

Click **Open**

- First time → click **Yes** (security alert)
- Login happens automatically as:

ubuntu

🍷 You are now connected to Ubuntu EC2

□ VERIFY CONNECTION

Run:

whoami

Output:

ubuntu

):

Connecting to EC2 Using AWS CloudShell

1. **Create the EC2 instance** using a key pair (.pem).

2. **Log in to AWS Console** and click on the **CloudShell** icon (terminal icon on the top-right).
 3. Once CloudShell opens, go to **AWS Services → EC2**.
 4. Select your **EC2 instance** and click **Connect**.
 5. Choose the **SSH client** tab.
 6. **Copy the ssh -i command** shown there.
 7. **Paste the command in CloudShell** and try to run it.
✗ It will fail with an error like:
"key pair is not available / no such file or directory"
(because the key file is not yet in CloudShell).
 8. Now, in **CloudShell**, click on the **"+" (Actions)** icon on the right side.
 9. Select **Upload file** and upload the **key pair (.pem)** from your local machine.
 10. After upload, **copy the ssh -i command again** from **EC2 → Connect → SSH client**.
 11. **Paste and execute the command** in CloudShell.
 12. ☒ You are now **successfully connected to the EC2 instance**.
-

Step-by-Step: Create Windows EC2 Instance and Connect via RDP

Step 1: Launch a Windows EC2 Instance

1. Log in to your **AWS Management Console**.
2. Navigate to **EC2 → Instances → Launch Instances**.
3. Select an **Amazon Machine Image (AMI)**:
 - Choose a **Windows Server** AMI (e.g., Windows Server 2019 or 2022).
4. Choose an **Instance Type** (e.g., t2.micro for free tier or as per your requirement).
5. Configure **Instance Details** as needed.
6. Add **Storage** if required (default is usually sufficient).
7. Configure **Security Group**:
 - Ensure **RDP port 3389** is allowed.

- Source can be **My IP** (for security) or a wider range if needed.

Firewall (security groups) | [Info](#)
 A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☒ Create security group
 ☐ Select existing security group

Security group name - *required*

This security group will be added to all network interfaces. The name can't be edited after the security group is created. Max length is 255 characters. Valid characters: a-z, A-Z, 0-9, spaces, and . _ - / () # , @ [] + = & ; ! \$ *

Description - *required* | [Info](#)

Inbound Security Group Rules

▼ Security group rule 1 (TCP, 3389, 152.57.149.255/32) Remove

Type Info	Protocol Info	Port range Info
rdp	TCP	3389

Source type Info	Name Info	Description - <i>optional</i> Info
My IP	Add CIDR, prefix list or security group 152.57.149.255/32	e.g. SSH for admin desktop

[Add security group rule](#)

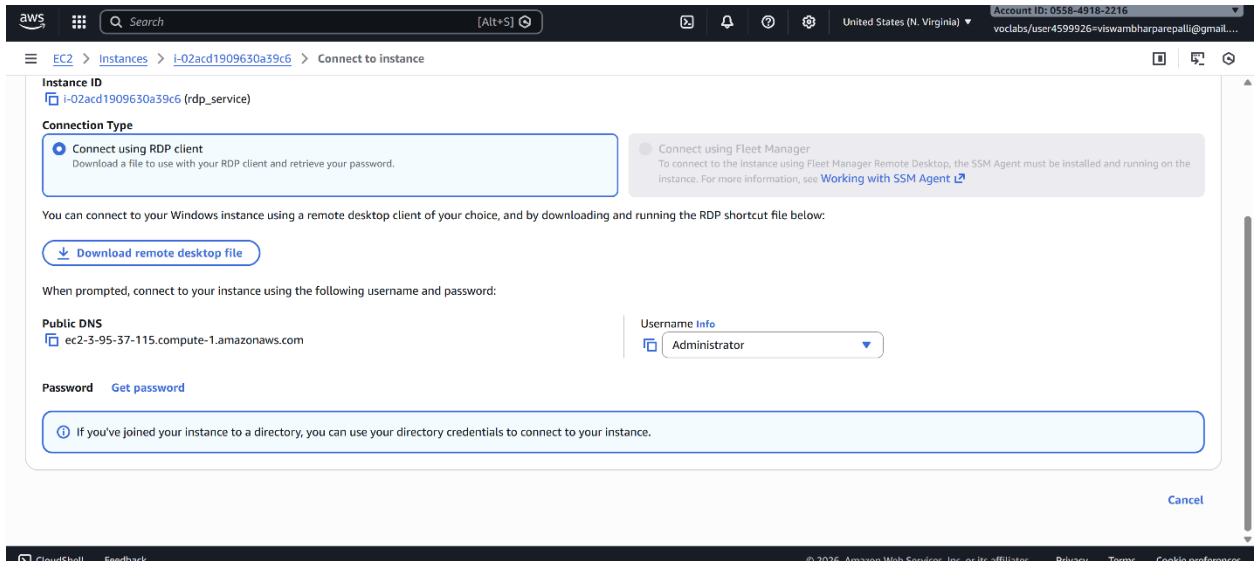
8. Select or **create a new Key Pair**:

- Download the .pem file (keep it safe; it's needed to decrypt the password).

9. Click **Launch Instance**.

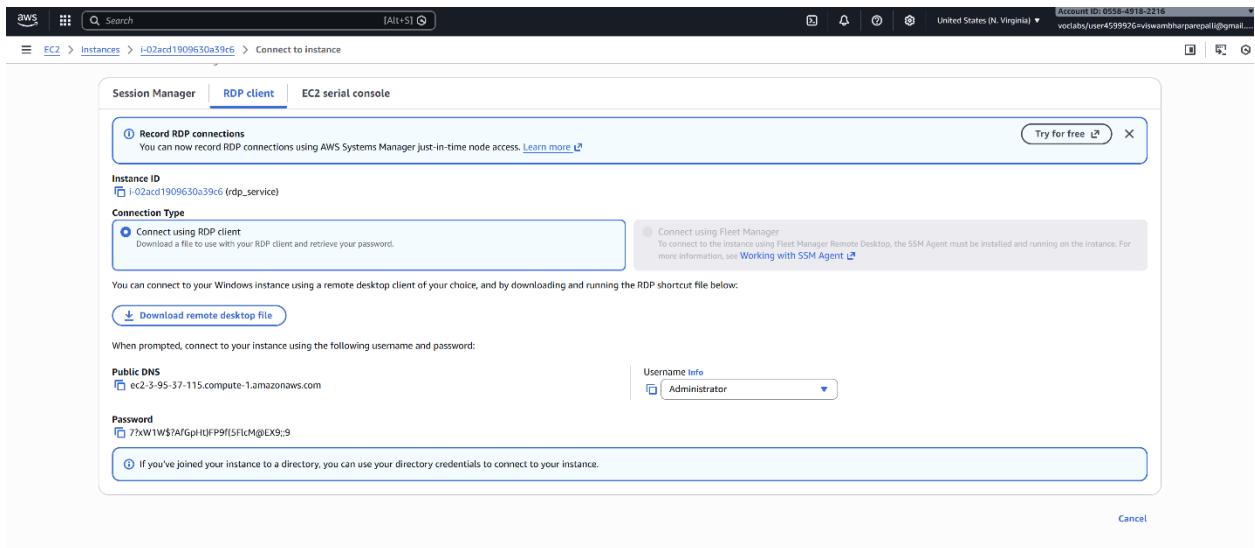
Step 2: Get the Public DNS / IP

1. After the instance is running, go to **EC2 → Instances**.
2. Select your instance and note the **Public DNS (IPv4)** or **Public IP**.



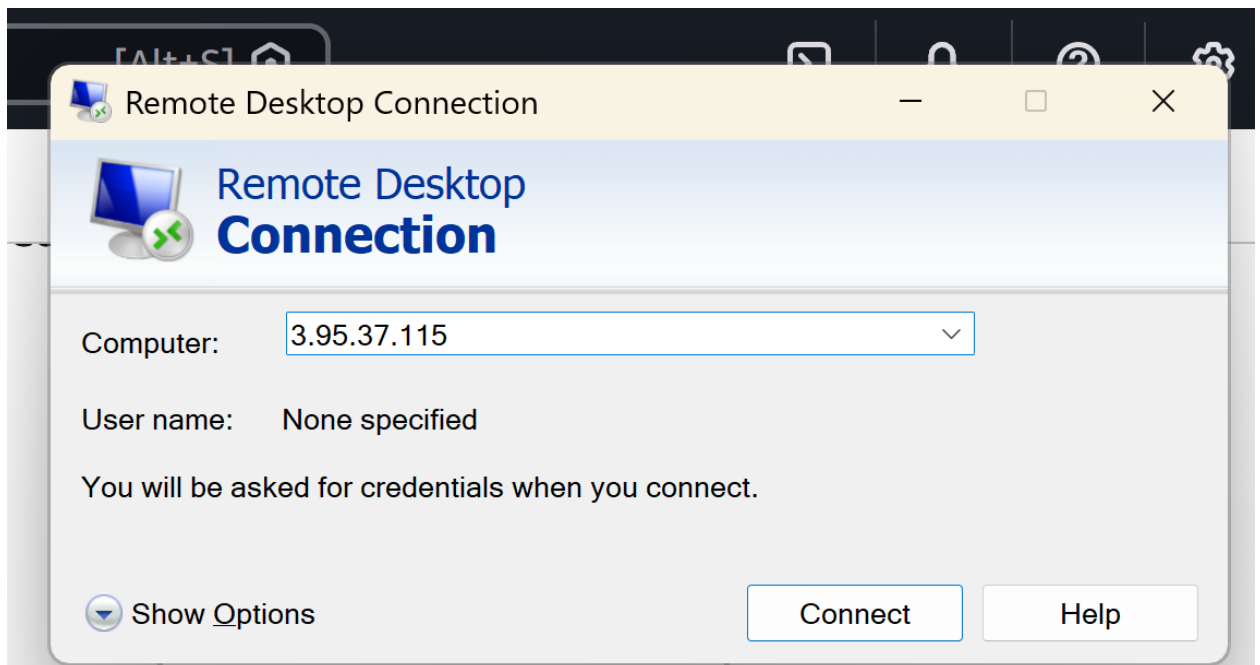
Step 3: Retrieve the Windows Administrator Password

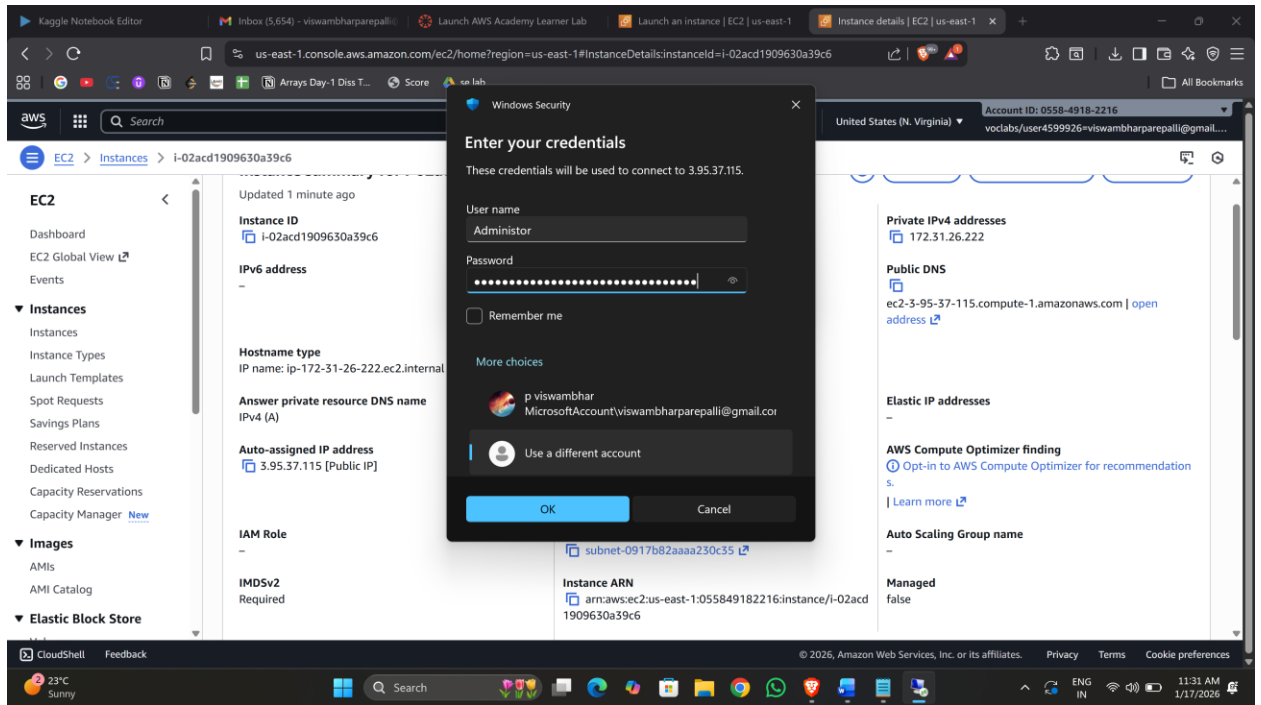
1. Select your instance → Click **Connect** → **RDP Client**.
2. Click **Get Password**.
3. Upload your **Key Pair (.pem)** file.
4. Click **Decrypt Password**.
5. Copy the **Administrator password** shown.



Step 4: Connect via RDP

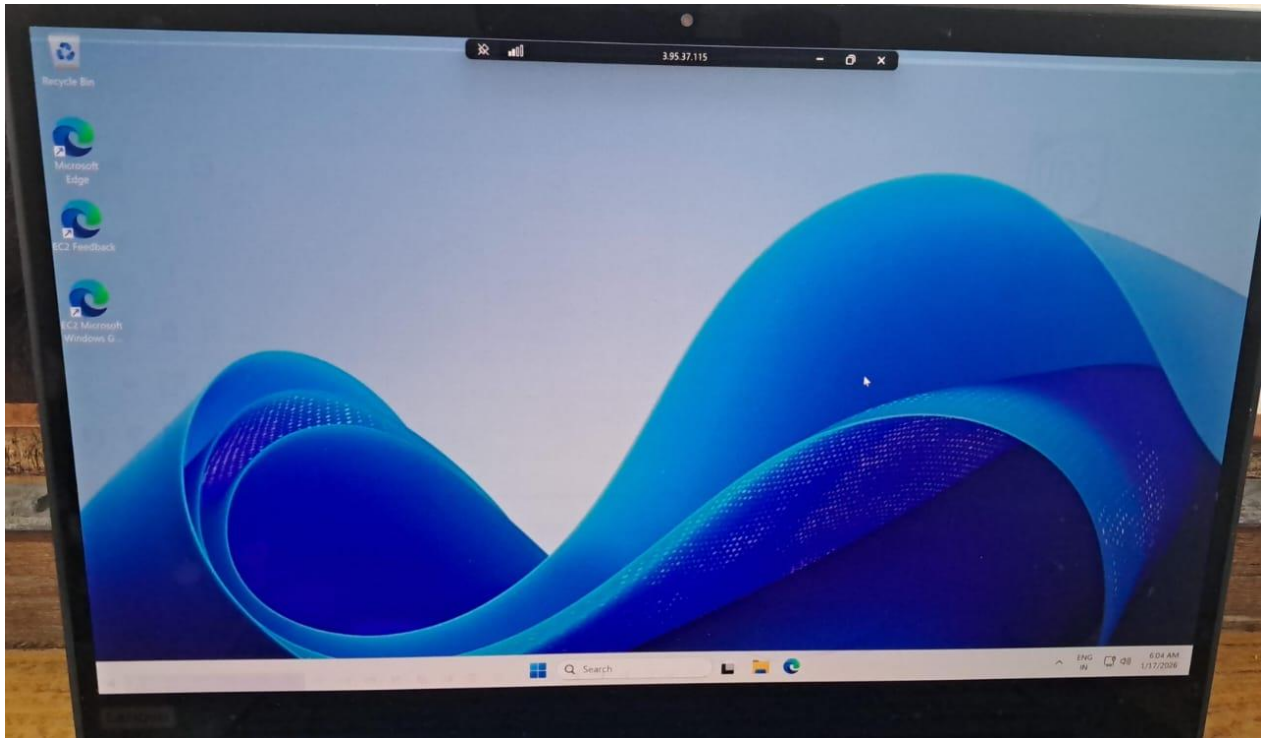
1. On your **local Windows machine**, open **Remote Desktop Connection** (mstsc).
2. In the **Computer** field, enter the **Public DNS** or **IP address** of the EC2 instance.
3. Click **Connect**.
4. In the login prompt:
 - **Username:** Administrator
 - **Password:** Paste the decrypted password from Step 3.
5. Click **OK**.





Step 5: You're connected

- Your Windows EC2 instance desktop should appear.
- You can now use it as a normal Windows machine.



VIVA QUESTIONS

Q1. What is AWS?

Amazon Web Services (AWS) is Amazon's cloud platform that lets you rent servers, storage, and other IT resources online instead of buying physical hardware. You pay only for what you use.

Q2. What are the key components of an EC2 instance?

An EC2 instance mainly includes: an OS image (AMI), instance type (CPU/RAM), storage (EBS), security groups (firewall), and key pairs (login access). Together, these define your virtual server.

Q3. How does EC2 work?

You choose a machine size and operating system, AWS launches a virtual server in minutes, and you connect to it over the internet. You can start, stop, resize, or delete it anytime, and billing runs only while it's active.

Q4. What are the uses, advantages, and examples?

Uses: hosting websites, running apps, testing software, data processing.

Advantages: no hardware cost, instant scaling, high reliability.

Examples: hosting a startup website, running backend APIs, creating dev/test servers, or training ML models.